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Validation of a Footwear-Based Gait Analysis System with Action-Related Feedback

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***Abstract*—Quantitative gait analysis enables clinicians to evaluate patient mobility and to diagnose a number of neuromuscular disorders. The clinical application of gait analysis is currently hampered by the high operating costs of gait laboratories. The use of instrumented footwear that performs out of the lab measurements on subject's walking pattern is a promising way to overcome this limitation. Besides serving as assessment tools, such devices can also act as retraining tools that help regulate a patient's gait by means of acoustic or vibrotactile stimuli.**

At the Columbia University Robotics and Rehabilitation (ROAR) Lab, we have developed SoleSound, a fully portable instrumented footwear that can measure spatiotemporal gait parameters and deliver action-related audio-tactile feedback. This paper presents the validation of SoleSound as a tool for quantitative gait analysis. Using a reference system, we assessed the performance of SoleSound under two calibration strategies, *subject-specific* and *generic*. With the latter calibration, mean accuracy and precision (RMSE±SD) were 2.93±1.32 cm for stride length, 0.70±0.37 cm for foot-ground clearance, 1.54±0.70 cm for step width, 4.53±0.90 cm for foot trajectory and 4.33±1.01 deg for ankle plantar-dorsiflexion angle. Accuracy was further improved when subject-specific calibration was used. These results indicate that SoleSound has the potential to be used for out-of-lab gait analysis

I. Introduction

Quantitative gait analysis - a method by which human locomotion is quantitatively characterized using equipment that captures kinematic and kinetic parameters - is a powerful tool for physicians treating patients with gait disorders. Gait analysis allows for more accurate diagnoses than those performed through observational evaluation and might lead to superior functional outcomes by informing treatment decision-making [1], [2]. Unfortunately, expensive laboratory equipment (namely, camera-based motion capture systems and instrumented walkway devices) and time-consuming data analysis lead to high procedure costs which many insurance providers do not cover; therefore, quantitative gait analysis is not widely used [2].

To overcome these limitations, researchers have recently developed wearable gait analysis devices, which consist of lightweight, battery-powered sensors worn on the lower body and a logic unit that processes and stores the collected data. These devices can be made for a fraction of the cost of current gait analysis systems and can be used in any location. They can be used to assess patients' gait remotely, over any distance or path (e.g., stairs climbing), and over extended time periods, thus enabling physicians to assess variability of gait or quantitatively monitor episodic phenomena, such as freezing of gait [3], that are difficult to elicit in clinical settings. Lastly, wearable devices can act as tools for gait retraining, providing auditory, vibrotactile or visual feedback to help the wearer correct movements when needed. However, these systems are limited in the number of gait parameters these can measure and in their accuracy [4]. Usually, these require a wired [5], [6] or wireless [7]–[10] connection to a host computer for online data processing or data-logging, which limits their actual mobility.

Wearable gait analysis devices can be classified based on the type of variables these measure: kinematic (spatial, temporal), or kinetic. Spatial parameters such as stride length, base of walking, foot clearance, foot trajectory and ankle range of motion (ROM) are powerful indicators of gait disorders. For example, widened walking base and short strides often indicate impaired balance [11]; reduced ankle ROM has been associated with higher risk of falls [12]–[14]; impaired control of foot trajectory during the swing phase might result in insufficient foot clearance that increases the chances of slipping or tripping [15]. Following the advent of miniaturized and relatively inexpensive inertial sensors, the last decade has seen a rapid increase in the number of research prototypes and off-the-shelf products intended for motion tracking, for pedestrian localization [16], [17] and for identification of posture and motor activities [18], [19]. Motion tracking devices are used in gait analysis or VR applications. These can be further classified into *inertial orientation trackers* - i.e., networks of inertial sensors that are worn on the subjec's limbs, including the lower extremities [10], [20]–[24], and optionally the trunk [25] and the arms [26] - and less obtrusive *shoe-based systems* that are worn on the foot [5], [9], [27], [28], and optionally on the shank [6], [8].

Spatial parameters are derived from position and orientation data measured by strap-down inertial units. In principle, 3D orientations relative to an inertial frame can be computed by integrating readings of a 3-axis rate gyroscope or by vector observation, with a 3-axis accelerometer providing inclination and a 3-axis magnetometer yielding the azimuth. Early wearable systems estimated sagittal plane joint angles by integrating the readings of single-axis gyroscopes located on each limb segment [20]. This approach, however, led to low frequency drift due to sensor bias. On the other hand, accelerometer observations are affected by movement dynamics

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and high frequency noise, and magnetometer observations suffer from distortion caused by local magnetic fields. Research in this field has focused on methods to improve accuracy of orientation estimations. Suggested approaches fall into three main categories: drift compensation based on gait timing information, sensor fusion algorithms applied to individual IMUs, and combination of sensor fusion with biomechanical constraints.

Shoe-based systems take advantage of the dynamics of walking, namely, the relatively stationary pose of the foot and shank during the foot flat period of stance (FF), to compensate sensors drift. Under pseudo-static conditions, dynamic contributions to the accelerometer readings are negligible, and the direction of gravity can be accurately tracked from these signals. FF phases are detected from on-board gyros [5], [8], [20], [21], force resistive sensors (FRS) [8], or multi-axis force sensors [6], [27], [29]. In planar models, gyroscope integration is conducted in between subsequent FF phases, the foot pitch angle is zeroed at the beginning of each phase, and linear drift compensation is applied by imposing zero pitch at the end of the integration interval [5], [8]. In spatial models, 3D orientation is adjusted with inclination estimates from a 3-axis accelerometer at any FF [27] or each time the norm of the acceleration is close to the gravity acceleration [9], [30]–[33].

A second approach relies on sensor fusion techniques. Extended Kalman Filters (EKF) have been widely used in inertial motor tracking. Several authors have proposed techniques to prevent spurious corrections from the EKF when pseudo-static conditions are not met. These include low-pass filtering of the acceleration observations [23], conditional correction based on the magnitude of the measured acceleration [34] and adaptive estimation of the measurement covariance matrix [35]. While the EKF is a powerful tool, it is also computationally intense and requires a-priori knowledge of the process dynamics. To overcome these shortcomings, researchers have proposed model-free filters, including the Weiner filter [25] and complementary filters [36].

The third approach seeks to further improve accuracy by taking advantage of both sensor fusion techniques and inherent biomechanical constraints. Cooper et al. [22] estimated the knee flexion/extension angle with two 6-DOF IMUs located on the thigh and shank. They modeled the knee as a simple revolute joint and assumed that the relative orientation between the two sensors at the initial time was known. Seel et al. [24] developed a similar method that does not rely on external calibration. Precision and accuracy reported in both works are superior to those yielded by previous approaches. Slajpah et al. [10] modeled the lower extremities as a serial kinematic chain, with the contact foot assumed to be a fixed point. They applied EKFs iteratively from the contact foot to the swinging foot, fusing the measured accelerations with these computed from the measured angular velocity of the link and the known kinematics of the previous links. A main drawback of their approach resides in the need for camera-based calibration of the kinematic model.

Many systems are capable of measuring the displacement of the foot over a gait cycle [6], [9], [29], [31]–[33], or the projection of the displacement onto the sagittal plane [5], [8], [31]. These estimates are then used to derive stride length and foot clearance. Although approaches based on biomechanical constraints have been proposed [20], the most widely used technique involves gravity cancellation from the foot accelerometer readings (by means of orientation estimates), followed by double integration [8]. Integration is commonly performed between subsequent FF phases, with velocity zeroing (ZUPT) and velocity drift compensation used to improve the accuracy of the estimates [5], [6], [9], [29], [31]–[33].

Temporal gait parameters such as stride time, stance, swing, double support periods, and temporal symmetry ratios [37] are usually estimated using in-shoe inertial sensors [5], [7], [20], [21]. A more direct approach, however, relies on sensing the underfoot forces through pairs of in-shoe force resistive sensors (FSR), from which the timing of heel strike and toe-off is derived [38]. More complex arrays of underfoot sensors have been proposed for accurate gait segmentation [39] as well as the estimation of kinetic parameters such as normal ground reaction forces (GRF) and the trajectory of the center of pressure (CoP) [6], [8], [34], [40]–[44].

Wearable motion tracking devices are capable of estimating spatiotemporal gait parameters, yet, *very few of them can also provide on-board feedback based on these estimates.* Head mounted displays have recently enabled wearable devices to provide visual feedback for gait and balance retraining [45], however, auditory and vibrotactile feedback has been more extensively explored in recent years. While simple discrete feedback (e.g., auditory alarms or trains of vibrations) has been successfully applied to balance or gait retraining [46]–[50], this strategy can convey only limited information to the wearer. Instead, advanced interactive sonification techniques - by which the output of auditory [51], [52], tactile [53] or audio-tactile [54] feedback engines is modulated by the user's gait - can produce action-related and ecologically-valid auditory or audio-tactile stimuli, which have been shown to affect the gait of healthy individuals [52], [55], [56], motivating further studies on gait retraining of neurological populations [57], [58]. These works indicate that wearable devices can be effectively used as gait retraining tools, reducing the risk of falls and helping subjects regulate gait and balance [45], [59].

Building on our group's previous experience with the PDShoe, a pair of water shoes outfitted with piezo-resistive sensors and vibrotactile motors that provide step-synchronized feedback [50], we have recently developed a new prototype of instrumented footwear named SoleSound [56]. Unlike similar devices, SoleSound is fully portable and does not require a host computer. Most importantly, it can measure a wide set of spatiotemporal gait parameters - including inter-limb variables such as the base of walking - and synthesize continuous audio-tactile feedback based on these. These *unique features* enable SoleSound to quantify the immediate and long-term effects of gait retraining with audio-tactile feedback without the need for laboratory equipment.

While the on-board audio-tactile feedback engine was discussed elsewhere [56], [60], the goal of this work is to experimentally assess accuracy and precision of gait analysis with SoleSound. After describing the latest design of the device (Sec. II-A) and how gait parameters are estimated

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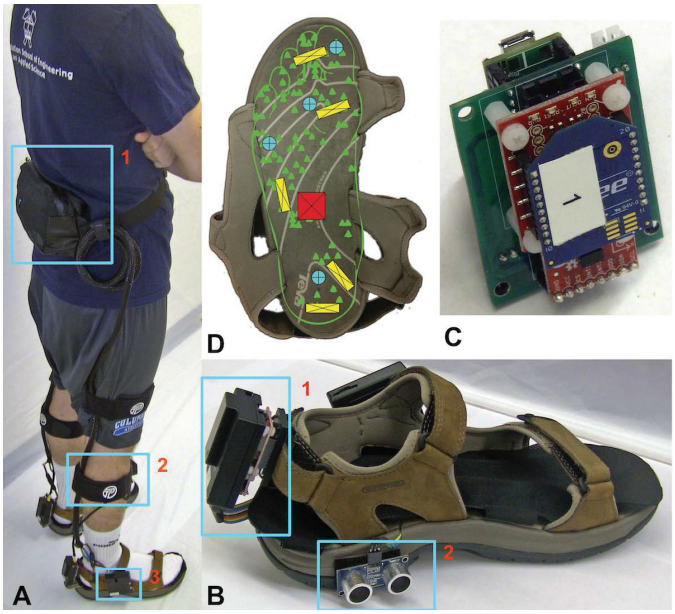


Fig. 1: (A) Subject wearing the SoleSound system; (B) footwear unit; (C) heel-mounted unit; (D) locations of the vibrotactile transducers (yellow rectangles) follow the average distribution of the cutaneous mechanoreceptors in the foot sole (green triangles, adapted from [61]); the locations of the pressure sensors are shown in cyan, while the red square indicates the position of the foot IMU.

(Sec. II-B), we illustrate the validation protocol and two distinct calibration approaches (Sec. II-C and II-D). Results are presented in Sec. III and discussed in Sec. IV.

II. Methods

A. Instrumented Footwear

SoleSound is a modular wearable system composed of two footwear units and a hip pack unit, Fig. 1-A. The current design revision has several significant improvements over the first prototype presented in [56], including faster sampling rate and the addition of an ultrasonic sensor and a shank IMU to estimate base of walking and ankle plantar-dorsiflexion angle, respectively. Each footwear unit measures pressure under the foot and kinematic data of the foot and shank. A real-time footstep synthesis engine running on a single-board computer processes these data and generates analog signals, which are amplified and fed to underfoot vibrotactile transducers and miniature loudspeakers mounted on the mid foot or to headphones, Fig. 2.

The foot module consists of a pair of sandals whose foam sole was cut to house five vibrotactile transducers (Tactile Labs, Montreal, Canada) and a 9-DOF IMU (Yost Engineering Inc., Portsmouth, OH). Two of the transducers were placed anteriorly (under the hallux and 1st metatarsal head), two posteriorly (under the calcaneus) and one underneath the lateral arch of the foot, approximately following the distribution of the cutaneous mechanoreceptors in the foot sole [61], Fig. 1-D. The inertial sensor was placed along the midline of the foot, below the tarsometatarsal articulations. Four piezo-resistive force sensors were secured to the sole of each shoe to measure the pressure underneath the calcaneus, the head

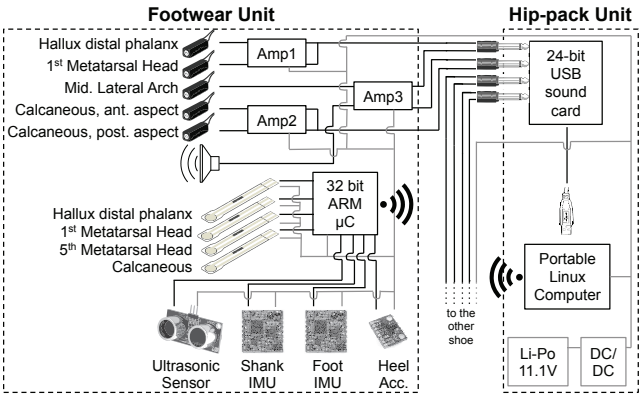


Fig. 2: System Layout

of the 5th metatarsal, the head of the 1st metatarsal, and the distal phalanx of the hallux. A second inertial unit is encased in a small plastic box secured to the user's proximal shank with Velcro, Fig. 1-A2. An ultrasonic sensor was mounted on the medial-posterior side of the sole to estimate the base of walking, Fig. 1-B2. Each sandal also features a custom-made driver box that is secured to the lateral posterior side of the footwear and contains three, 2-channel audio amplifier boards to power the transducers, Fig. 1-A3. When headphones are not used, a miniature loudspeaker optionally attaches to the anterior strap of the sandal and can be directly powered from the driver box.

The control logic responsible for acquiring the signals from the sensors runs on a 32-bit ARM Cortex-M4 Processor. This unit was encased on a heel-mounted box, along with a 3-axis accelerometer and a Wi-Fi antenna, Fig. 1-B1. Data sampled from each foot unit are sent through UDP over WLAN to a portable single-board computer, which synchronizes the data, runs the feedback engine and performs data-logging to a micro SDcard at 500Hz. The hip pack also houses an external 24-bit sound card (ASUSTeK Computer Inc.), a small Wi-Fi router, and a Li-Po battery, Fig. 1-A1. Feedback is generated in real-time by the audio-tactile footstep synthesis engine described in [56], [60], and is subsequently converted to eight independent analog signals - four per leg - by the sound card. A pair of thin stereo audio cables bundled in PET cable sleeving attaches to the wearer's thighs and shanks, carrying the analog signals from the waist to the driver box mounted on each foot. A user interface implemented in Matlab (The MathWorks, Natick, MA, USA) allows the experimenter to wirelessly connect to the on-board controller, monitor in real-time the data measured by the sensors and tune the parameters of the feedback engine if necessary.

Unlike existing devices, which feature either wired or wireless connections between the footwear and the central controller, SoundSole utilizes a hybrid configuration, namely, sensor data are sent from the footwear to the hip pack unit wirelessly, while feedback signals are delivered to the transducers through cables. The wireless network solution adopted for the input data makes the SoleSound a modular system: additional input sources (e.g., inertial sensors on the waist or upper body) can be added with minimal additional

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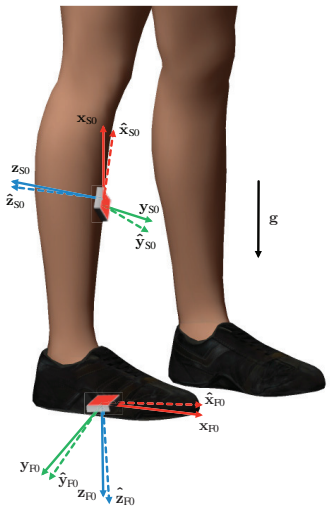


Fig. 3: Reference frame and reference tare frame for the foot and shank IMU.

programming. The 500Hz sample rate and the tethered connection for the feedback contribute to reduce the overall latency (sensor inputs to motor output) to less than 35 ms. The full system can be donned in about 5 minutes without assistance; the total weight of the components attached to each sandal is 0.19 kg and that of the hip pack module is 1.14 kg.

B. Spatio-temporal Gait Analysis

SoleSound is capable of estimating temporal and spatial gait parameters. The use of FRS to estimate temporal gait parameters is common practice [38] and therefore was not object of validation. Instead, accuracy and precision of spatial parameters were assessed. These comprise ankle plantar-dorsiflexion angle (including ankle range of motion, or ROM, and ankle symmetry), foot trajectory (including stride length and foot-ground clearance) and step width.

Each foot and shank IMU provide orientation estimation relative to a reference (tare) frame based on an on-board proprietary EKF algorithm that weights the contributions of the accelerometer and magnetometer based on the current dynamics experienced by the inertial sensor within a user-selectable range of feasible weights [62]. The foot IMU is embedded in the sandal sole, with the local axis $\hat{\mathbf{z}}_{\mathrm{F}}$ orthogonal to the sole and pointing downward and the local axis $\hat{\mathbf{x}}_{\mathrm{F}}$ aligned with the longitudinal axis of the sandal, Fig. 3. At startup, the subject is asked to stand stationary for 5 seconds and the reference orientations for the foot and shank IMUSs are set as described below. The mean acceleration values measured in the startup interval define the direction of the gravity vector $\mathbf{g}$ relative to the local IMU frames of foot and shank. The reference frame of the foot $\{\mathrm{F0}\}$ is defined as

$$\mathbf{z}_{\mathrm{F0}} = \mathbf{g}, \quad \mathbf{x}_{\mathrm{F0}} = \frac{\hat{\mathbf{x}}_{\mathrm{F0}} - (\hat{\mathbf{x}}_{\mathrm{F0}} \cdot \mathbf{z}_{\mathrm{F0}})\mathbf{z}_{\mathrm{F0}}}{\|\hat{\mathbf{x}}_{\mathrm{F0}} - (\hat{\mathbf{x}}_{\mathrm{F0}} \cdot \mathbf{z}_{\mathrm{F0}})\mathbf{z}_{\mathrm{F0}}\|}, \quad \mathbf{y}_{\mathrm{F0}} = \mathbf{z}_{\mathrm{F0}} \times \mathbf{x}_{\mathrm{F0}}, \quad (1)$$

where $\hat{\mathbf{x}}_{\mathrm{F0}}$ is the local axis $\hat{\mathbf{x}}_{\mathrm{F}}$ at $t = 0$. The shank IMU is attached to the subject's proximal shank with a Velcro wrap. The local axis $\hat{\mathbf{x}}_{\mathrm{S}}$ is assumed to be aligned with the longitudinal axis of the tibia, pointing upward, and the local axis $\hat{\mathbf{z}}_{\mathrm{S}}$ is directed posteriorly, Fig. 3. Similar to the foot, the reference frame of the shank $\{\mathrm{S0}\}$ is defined as

$$\mathbf{x}_{\mathrm{S0}} = -\mathbf{g}, \quad \mathbf{z}_{\mathrm{S0}} = \frac{\hat{\mathbf{z}}_{\mathrm{S0}} - (\hat{\mathbf{z}}_{\mathrm{S0}} \cdot \mathbf{x}_{\mathrm{S0}})\mathbf{x}_{\mathrm{S0}}}{\|\hat{\mathbf{z}}_{\mathrm{S0}} - (\hat{\mathbf{z}}_{\mathrm{S0}} \cdot \mathbf{x}_{\mathrm{S0}})\mathbf{x}_{\mathrm{S0}}\|}, \quad \mathbf{y}_{\mathrm{S0}} = \mathbf{z}_{\mathrm{S0}} \times \mathbf{x}_{\mathrm{S0}}, \quad (2)$$

with $\hat{\mathbf{z}}_{\mathrm{S0}}$ being the local axis $\hat{\mathbf{z}}_{\mathrm{S}}$ at $t = 0$. Assuming neutral subtalar position and neutral knee alignment during the taring process, the mapping between $\{\mathrm{F0}\}$ and $\{\mathrm{S0}\}$ is given by the following anti-diagonal matrix:

$${}^{\mathrm{S0}}_{\mathrm{F0}}\mathrm{R} = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix} \quad (3)$$

For $t > 0$, the orientation estimations of foot and shank relative to their respective reference frames are returned in terms of yaw-pitch-roll Euler angles. Together with (3), these data are sufficient to derive the three ankle angles: abduction/adduction, inversion/eversion and plantar/dorsiflexion. Here, we focus on the ankle plantar/dorsiflexion angle γ_{PD}, because it is the most critical for gait propulsion and support against gravity [63]. γ_{PD} is defined as the relative pitch angle between foot and shank, offset by $\pi/2$ [12]. As shown by (3), the axes $\mathbf{y}_{\mathrm{S0}}$ and $\mathbf{y}_{\mathrm{F0}}$ are antiparallel, yielding

$$\gamma_{\mathrm{PD}} = \theta_{\mathrm{F}} + \theta_{\mathrm{S}}, \quad (4)$$

where θ_{F} and θ_{S} are the pitch angles of the foot and shank, respectively. For each leg, the ankle angle (4) is segmented into gait cycles (GC) using the readings of the heel pressure sensors as detectors of initial contact (IC). For the i-th stride of each leg, the ankle angle is then time-normalized over the GC and downsampled into N equally spaced points to yield the ankle trajectory $\bar{\gamma}_{\mathrm{PD\,i}}$. The ankle range of motion $\mathrm{ROM_i}$ is defined as the difference between the absolute maximum and minimum of $\bar{\gamma}_{\mathrm{PD\,i}}$. A gait symmetry metric $\mathrm{SYM_i}$ is derived as the RMS deviation between the normalized ankle trajectories of the right and left legs, corresponding to two consecutive strides

$$\mathrm{SYM_i} = \sqrt{\frac{\sum_{j=1}^{N} (\bar{\gamma}_{\mathrm{PD_LEFT\,i,\,j}} - \bar{\gamma}_{\mathrm{PD_RIGHT\,i,\,j}})^2}{N}}, \quad (5)$$

with N being the number of samples in $\bar{\gamma}_{\mathrm{PD\,i}}$.

The foot IMU returns the components of the acceleration vector $\mathbf{a}$ (deprived of the gravity component) in the reference frame $\{\mathrm{F0}\}$. A threshold-based algorithm detects the FF period as the fraction of the stance phase wherein the Euclidean norm of $\mathbf{a}$ is smaller than a predefined threshold. First, the foot velocity in the i-th stride $\mathbf{v}_{\mathrm{i}}$ is obtained by integration of $\mathbf{a}$, with the medians of the i-th and $(i+1)$-th FF periods defining the i-th interval of integration [9]:

$$\mathbf{v}_{\mathrm{i,\,j}} = \mathbf{v}_{0\mathrm{i}} + \frac{1}{f_s} \sum_{k=\mathrm{FF}_i}^{\mathrm{FF}_i + j - 1} \mathbf{a}_k, \quad j \in [1,\ \mathrm{FF_{i+1}} - \mathrm{FF_i} + 1], \quad (6)$$

$\mathbf{v}_{\mathrm{i,\,j}}$ is the linear velocity of the foot in the j-th sample of the i-th stride, $[\mathrm{FF_i}, \mathrm{FF_{i+1}}]$ is the interval of integration for the i-th stride and f_s is the sampling frequency. The constant of integration $\mathbf{v}_{0\mathrm{i}}$ is set to zero (ZUPT technique) and the raw velocity estimate (6) is corrected to compensate for velocity drift (assumed linear):

$$\bar{\mathbf{v}}_{\mathrm{i,\,j}} = \mathbf{v}_{\mathrm{i,\,j}} - \frac{j-1}{\mathrm{FF_{i+1}} - \mathrm{FF_i}} \mathbf{v}_{\mathrm{i,\ FF_{i+1}} - \mathrm{FF_i} + 1} \quad (7)$$

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Fig. 4: Positions of the reflective markers on a subject

The foot displacement $\mathbf{d}_{\mathrm{i}}$ is computed by integration of $\bar{\mathbf{v}}_{\mathrm{i}}$:

$$\mathbf{d}_{\mathrm{i,\,j}} = \frac{1}{f_s}\sum_{k=1}^{j}\bar{\mathbf{v}}_{\mathrm{i,\,j}}\,, \quad j \in [1,\,\mathrm{FF_{i+1}}\text{ - }\mathrm{FF_i}+1]\,, \tag{8}$$

where $\mathbf{d}_{\mathrm{i,\,j}}$ is the displacement of the foot in the j-th sample of the i-th stride. $\mathbf{d}_{\mathrm{i}}$ is known in $\{\mathrm{F0}\}$, however, for the purposes of gait analysis, the reference frame $\{\mathrm{Di}\}$ aligned with the direction of progression is more desirable:

$$\mathbf{x}_{\mathrm{Di}} = \frac{\mathbf{d}_{\mathrm{i,\,FF_{i+1}\text{ - }FF_i+1}} - \mathbf{d}_{\mathrm{i,\,1}}}{\left\|\mathbf{d}_{\mathrm{i,\,FF_{i+1}\text{ - }FF_i+1}} - \mathbf{d}_{\mathrm{i,\,1}}\right\|}, \quad \mathbf{z}_{\mathrm{Di}} = -\frac{\mathbf{z}_{\mathrm{F0}} - (\mathbf{z}_{\mathrm{F0}}\cdot\mathbf{x}_{\mathrm{Di}})\,\mathbf{x}_{\mathrm{Di}}}{\left\|\mathbf{z}_{\mathrm{F0}} - (\mathbf{z}_{\mathrm{F0}}\cdot\mathbf{x}_{\mathrm{Di}})\,\mathbf{x}_{\mathrm{Di}}\right\|}, \quad \mathbf{y}_{\mathrm{Di}} = \mathbf{z}_{\mathrm{Di}}\times\mathbf{x}_{\mathrm{Di}} \tag{9}$$

$\bar{\mathbf{d}}_{\mathrm{i}}$ - the sagittal-plane, normalized foot trajectory for the i-th stride - is obtained by projecting $\mathbf{d}_{\mathrm{i}}$ onto the $\mathbf{x}_{\mathrm{Di}}\mathbf{z}_{\mathrm{Di}}$ plane, time-normalizing over the interval $[1,\,\mathrm{FF_{i+1}}\text{ - }\mathrm{FF_i}+1]$, and downsampling into N equally-spaced points. Finally, stride length $\mathrm{SL_i}$ and foot ground clearance $\mathrm{SH_i}$ are defined as

$$\mathrm{SL_i} = \left|\bar{\mathbf{d}}_{\mathrm{i,\,N}}(x) - \bar{\mathbf{d}}_{\mathrm{i,\,1}}(x)\right|, \quad \mathrm{SH_i} = \max_{\mathrm{j}\,\in\,[1,\,\mathrm{N}]}\left(\bar{\mathbf{d}}_{\mathrm{i,\,j}}(z)\right), \tag{10}$$

with $\bar{\mathbf{d}}_{\mathrm{i,\,j}}(x)$ and $\bar{\mathbf{d}}_{\mathrm{i,\,j}}(z)$ being the projections of $\bar{\mathbf{d}}_{\mathrm{i,\,j}}$ onto $\mathbf{x}_{\mathrm{Di}}$ and $\mathbf{z}_{\mathrm{Di}}$, respectively.

In gait analysis, there is no clear consensus on the definition of step width [64]. In this work, we estimated step width as the foot separation at mid-swing [65]. During overground walking in a straight-line, the ultrasonic sensor mounted on the medial posterior site of the left sole returns a minimal distance when the forward swinging left foot passes the stance foot. The step width of the i-th stride $\mathrm{SW_i}$ is therefore estimated by the absolute minimum of the ultrasonic sensor readings during the swing phase of the i-th left stride.

C. Experimental Protocol

The raw metrics described in the previous section are affected by systematic and random errors. Not only can these errors be quantified experimentally by comparison with the data collected by a laboratory-grade motion capture system (regarded as the *ground truth*), but the same data can also be used to calibrate the less accurate system, largely compensating for the systematic errors and thereby improving the level of agreement between the two gait analysis systems. To this end, data were collected from fourteen healthy adult individuals with no gait abnormalities (10 males, 4 females, age 26.6±4.2 years, height 1.70±0.10 m, weight 64.9±9.5 kg, US shoe size 8.0± 2.5). Experiments consisted of a single visit to the laboratory. The Columbia University Institutional Review Board approved the protocol for this study, and all subjects provided informed consent prior to the experimental session.

Subjects chose an appropriate shoe size and SoleSound was attached as shown in Fig. 1-A. Reflective markers were placed on both legs, either on anatomical landmarks (medial and lateral malleoli and femoral condyles, distal and proximal tibia) or on the sandals (close to the hallux, the calcaneus, and the heads of the 1st, 2nd and 5th metatarsal), Fig. 4. The centroid of the markers attached to each sandal was regarded as the reference point to estimate foot displacement. Prior to the test, subjects were asked to stand stationary for 5 seconds. The on-board inertial sensors were zeroed at this time, and *never reset during the session*. Participants were instructed to complete 30 laps at a self-selected, comfortable pace. During each lap, they walked along a 14 m long, straight-line path marked on the floor, made a clockwise turn, and went back to the starting point. Each session lasted approximately 15 minutes. Subjects' movements were simultaneously recorded by SoleSound and a camera-based motion capture system with 10 cameras (Vicon Bonita B10, Oxford, UK). Sample rates were set to 500Hz and 100Hz, respectively, and an infrared LED controlled by SoleSound was used to sync the two systems. A 5 m section in the middle of the first leg of each lap was regarded as representative of steady state walking, and the corresponding strides were included in the analysis described below.

D. Data Analysis

Gait parameters estimated by SoleSound can be divided into *scalar* parameters (i.e., $N = 1$ sample per stride) and *vector* parameters (i.e., $N = 101$ samples per stride, uniformly distributed in the interval 0-100% GC). Stride length (SL), foot ground clearance (SH), base of walking (SW), ankle symmetry (SYM) and ankle range of motion (ROM) belong to the first group. Vector parameters include ankle angle (γ_{PD}) and foot trajectory ($\bar{\mathbf{d}} = \left[\bar{\mathbf{d}}(x)\;\bar{\mathbf{d}}(z)\right]$). The calibration approach described below applies to both groups.

The raw metrics from SoleSound and the data from the camera-based system were processed using custom MATLAB code (The Mathworks, Natick, MA, USA). Prior to the actual calibration, an optimization script was implemented to determine the order and the cutoff frequency of the low-pass Butterworth filter applied to the norm of the foot acceleration $\|\mathbf{a}\|$, as well as the optimal threshold used to estimate FF periods from the measured acceleration. This optimization was based exclusively on raw training data, and yielded a 4th order filter with 8 Hz cutoff frequency, and a threshold of 0.15 m/s^2 Then, two alternative calibration approaches were implemented as described in the following.

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1) Subject-specific calibration: this calibration requires training datasets of a specific participant and outputs a set of calibration coefficients that are tailored to that subject. For the purposes of this experiment, the training datasets p_{tr}^V and p_{tr}^S (where the superscripts V and S indicate the reference system and SoleSound, respectively) were obtained for each subject and each parameter by selecting every other stride from the full set of data, while the remaining data formed the testing datasets p_{ts}^V and p_{ts}^S. In practice, this approach guarantees higher accuracy at the cost of a longer setup time, as it requires the use of a reference system prior to testing. Once the calibration has been completed inside a gait laboratory, the subject's gait can be tested in the unconstrained environment (stairs climbing and descending, ramps, etc.). For each parameter p, N linear regression models were generated in the form [66]

$$p_{tr}^V(j) \sim p_{tr}^S(j), \quad j \in [1, N], \tag{11}$$

where $p_{tr}^*(j)$ is the j-th sample of p measured by SoleSound or by the reference system. These models yielded $\beta_{0,j}$ and $\beta_{1,j}$, the optimal coefficients (in the least square sense) which minimize the sum of the squared residuals. The estimate of p at the i-th stride was computed as

$$\hat{p}_i^S(j) = \beta_{0,j} + \beta_{1,j}\ p_{ts,i}^S(j), \quad j \in [1, N], \tag{12}$$

and the associated error was calculated as:

$$e_i(j) = \hat{p}_i^S(j) - p_{ts,i}^V(j), \quad j \in [1, N] \tag{13}$$

This approach was independently applied to each subject's dataset.

2) Generic calibration: for each subject, the calibration coefficients were computed based on the data sampled from all the other subjects (training dataset), while the excluded subject's data formed the testing dataset (*leave-one-out cross validation*, or LOOCV [67]). In practice, this type of calibration is representative of the general application of SoleSound, when it is unpractical or unfeasible to perform a subject-specific calibration prior to using the system. In this case, the basic linear model was augmented with the subjects' anthropometric characteristics listed below:

$$p_{tr}^V(j) \sim p_{tr}^S(j) + \text{Height} + \text{Weight} + \text{Shoe Size} + \text{Age} + \text{Gender},\ j \in [1, N] \tag{14}$$

Solving the least square problem yielded $m + 2$ regression coefficients ($\beta_0 \dots \beta_{m+1}$), with $m = 5$ being the number of anthropometric characteristics included in the model. Models with $m < 5$ were also implemented and compared to the full model (14) with an ANOVA. The analysis revealed that simpler models yielded significantly worse results. The estimate of p at the i-th stride was computed as

$$\hat{p}_i^S(j) = \beta_{0,j} + \beta_{1,j}\ p_{ts,i}^S(j) + \sum_{k=1}^{m} \beta_{k+1,j}\ x_k, \quad j \in [1, N], \tag{15}$$

where x_k is the covariate related to the k-th anthropometric characteristic. The measurement error was calculated with (13). This procedure was iterated 14 times, once for each subject.

TABLE I: Results (mean RMSE $\pm$ SD)

		Subject specific	Generic
Ankle ROM [deg]	ROM	2.12 $\pm$ 0.63	4.76 $\pm$ 1.91
Ankle Symmetry [deg]	SYM	1.95 $\pm$ 0.38	2.72 $\pm$ 1.53
Stride Length [cm]	SL	2.30 $\pm$ 0.90	2.93 $\pm$ 1.32
Foot-Ground Clearance [cm]	SH	0.38 $\pm$ 0.10	0.70 $\pm$ 0.37
Base of Walking [cm]	SW	0.82 $\pm$ 0.19	1.54 $\pm$ 0.70
Ankle Angle [deg]	γ_{PD}	2.70 $\pm$ 0.39	4.33 $\pm$ 1.01
Foot Trajectory [cm]	$\bar{\mathbf{d}}$	3.30 $\pm$ 0.32	4.53 $\pm$ 0.90

III. Results

A total of 1888 strides was acquired by SoleSound and by the reference system (i.e., 4-5 gait cycles for each of the 30 laps, for each subject). Subjects' walking speed varied between 0.74 m/s and 1.47 m/s (1.14 $\pm$ 0.20 m/s, mean $\pm$ SD). Results are reported in Tab. I in terms of (mean RMSE $\pm$ SD) for both calibration strategies. Fig. 5 shows the correlation plots between SoleSound and the reference system (Fig. 5(a)-(f)), the frequency distribution of the measurement error (Fig. 5(g)-(l)) and the Bland-Altman plots [68] (Fig. 5(m)-(r)) for a subset of the scalar parameters.

Fig. 5(s)-(t) shows the ankle dorsiflexion angle averaged across all subjects, and Fig. 5(u)-(v) illustrate the average foot trajectory for a representative subject. Shaded areas indicate +/- 1 SD.

Several related studies reported the performances of wearable devices in terms of accuracy and precision (mean error $\pm$ SD) rather than in terms of RMSE. This alternative convention is directly related to the diagrams shown in Fig. 5(g)-(l). Under this convention, the results reported in Tab. I translate as: 0.27 $\pm$ 2.40 cm for SL, -0.01 $\pm$ 0.39 cm for SH, -0.01 $\pm$ 0.84 cm for SW in the case of the subject-specific calibration. The corresponding values for the generic calibration are: 0.01 $\pm$ 3.28 cm for SL, 0.06 $\pm$ 0.79 cm for SH, and -0.30 $\pm$ 1.65 cm for SW.

IV. Discussion

The study presented in this paper experimentally validated the accuracy and precision of SoleSound, the first *fully portable* instrumented footwear capable of delivering continuous auditory and vibrotactile underfoot feedback to the wearer by means of a real-time feedback engine informed by the readings of on-board inertial and piezoresistive sensors. SoleSound is capable of measuring a more complete set of gait parameters than any previously described shoe-based device. Furthermore, most shoe-based gait retraining systems do not provide sensing and feedback on-board, but require tethered connections to laboratory equipment. Complete portability is a key advantage of SoleSound: the device can perform gait analysis outside specialized laboratories and without the need for expensive motion capture systems.

SoleSound measures two types of gait parameters: spatial parameters, which include stride length, foot-ground clearance, base of walking, foot trajectory, and ankle plantar-dorsiflexion angle; and temporal parameters which include cadence, single/double support, symmetry ratios, and walking speed. Wireless communication and data logging are performed at 500

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Hz, a sampling frequency higher than any reported by related works, thus reducing latency in the sound feedback.

To make SoleSound more practical for clinical applications, complex alignment procedures were not adopted. Precise alignment of IMUs and anatomical segments usually requires preliminary calibration steps, which can be accomplished either with custom-made jigs [25] or with a camera based motion capture system, by rigidly attaching a cluster of reflective markers to the mounting plate of each inertial sensor [9], [10], [22], [29]. These steps must be completed prior to each experimental session to guarantee the level of accuracy reported. Such methods reduce the portability of the wearable system. Even though algorithms that automatically estimate sensor locations with respect to the anatomical axes based on kinematic constraints have been recently proposed [24], these assume that the relative axes of rotation of the human joints are fixed with respect to body segments, an approximation that affects their accuracy [69]. In the calibration method presented here, instead, markers were placed exclusively on anatomical landmarks (as it is common practice in quantitative gait analysis), thus making the reported results independent of precise alignment of the IMUs to the human limbs.

Instead of relying on professional-grade inertial sensors to improve the system's performance, SoleSound achieved the same target using mid-grade, cost-effective IMUs, by adopting linear calibration techniques. After deriving linear models based on raw datasets and corresponding reference datasets (as discussed in Sec. II-D), linear corrections were successfully used to reduce systematic errors. Even though calculation of the linear models had to be carried out off-line, applying the models requires minimal computational cost, and is therefore suitable for real-time applications using micro-controllers. In the generic calibration, *leave-one-subject-out cross-validation* was used to estimate the inter-subject variability, an approach that is fairly common in the machine learning community, but less popular in the biomechanics community. To further improve accuracy and precision, a subject-specific calibration was proposed.

The estimates of stride length, foot ground clearance and base of walking demonstrated a level of agreement in line with similar shoe-based devices, as indicated by the Bland-Altman plots (Fig. 5(m)-(r)). For the stride length, better results were obtained in terms of accuracy and precision compared to similar shoe-based systems [8], [32]. With the subject-specific calibration, SoleSound yielded better accuracy than those reported by Mariani et al. [33], Schepers et al. [29] and Weenk et al. [70] while in terms of precision, performance was second to Schepers' device only. Notably, each of these works used professional-grade IMUs. With the generic calibration, precision degraded below the values reported in [33] and [29], but still outperformed the one presented in [70], while in terms of accuracy, these comparisons were reversed. Estimates of the foot ground clearance outperform those reported by previous works [9], [31]. The average RMSE for ankle dorsiflexion angle obtained with SoleSound is lower than what reported in [21], but slightly higher than what presented in [10], [24]. The approach suggested by Slajpah et al. [10], however, relies on the knowledge of the position and orientation of the inertial sensors relative to each limb, while the results reported by Seel et al. [24] - although promising - do not account for inter-subject variability, since their method was validated on a single subject who performed only six short walking trials.

Few authors assessed the accuracy of foot trajectory estimates made by wearable devices during walking. Sabatini et al. [5] computed the trajectory of a reference point of the foot on the sagittal plane, but their results were not validated against a reference system. Schepers and colleagues [6] estimated the 3D position of the heel: their results outperform the ones obtained with SoleSound; however, they did not account for intra and inter-subject variability because only 40 strides from a single subject were analyzed. Morris-Bamberg et al. [8] estimated the foot displacements during walking, but this analysis was limited to one dimension (i.e., the line of progression). The RMSE on the estimation of the foot trajectory obtained with SoleSound are deemed acceptable, being smaller than 2.5% SL and 3.5% SL for the subject-specific calibration and the generic calibration, respectively.

The capability of measuring the base of walking and spatiotemporal gait symmetry are additional novel contributions of this work. The base of walking has rarely been studied by designers of shoe-based devices, although it is a key inter-limb parameter to assess the user's balance. A similar capability was claimed in [8], but the authors did not present quantitative data to support their assertion. The device introduced in [29] estimated the lateral position of one foot w.r.t. the line of progression: a unilateral metric that is correlated, but not equivalent, to the base of walking.

Weenk et al. [70] recently presented an EKF for shoe-based devices that fuses data from two on-board IMUs and two ultrasonic transducers to estimate stride width and length. Although simpler, the SoleSound approach yielded higher precision and accuracy than Weenk's method in the subject specific calibration (0.66 ± 0.15 cm when converted in terms of mean absolute error, or MAE) and more precise but slightly less accurate estimates in the the generic calibration (1.34 ± 0.69 cm in terms of MAE). Further, it must be noted that the method described in [70] was tested only off-line, and on a limited sample of steps collected from three subjects.

SoleSound is capable of estimating the base of walking in a relatively accurate way, although the accuracy decreases for high values of the stride width, especially in the generic calibration. This could be due to the sensitive range of the ultrasonic sensor.

Overall, the results obtained with SoleSound provide further evidence that shoe-based systems could represent a valid alternative to specialized laboratory equipment for quantitative analysis of a basic set of spatiotemporal gait parameters.

Some design features of SoleSound can be improved. The sonar will be replaced with a more compact and accurate model, and the driver box will be further miniaturized. Using a single-board computer with built-in wireless capability will help reduce the weight of the hip-pack unit.

In the experimental validation, we focused on straight-line walking. A more robust validation should include other conditions, such as stair-climbing and walking on a figure-eight pathway [9], [10]. Another limitation of the current

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experimental protocol is the relatively short duration of the walking trials (15 min). Even though this duration is significantly longer than those included in most related studies [6], [9], [22], longer duration would enable the use of SoleSound in long-term training protocols. It would also be advisable to further check the reliability of the subject-specific calibration with a test-retest experimental design. Lastly, the proof-of-concept validation of SoleSound presented in this work was carried out on healthy young subjects, even though elderly individuals and patients with central neurological disorders are its envisioned target populations. Therefore, future studies on clinical population will include separate calibrations. These studies will address the immediate and long-term effects of gait training with auditory and vibro-tactile underfoot feedback on the gait of elderly or neurologically impaired individuals.

V. Conclusion

Low-cost wearable technologies for clinical applications recently gained increasing interest from the research community. In this work, we presented the design and validation of SoleSound, a novel, wearable shoe-based gait analysis system, which can further deliver auditory and vibro-tactile feedback in response to measured gait parameters to help the user regulate her gait. While the feedback engine and its immediate effect on the gait of healthy users were described in previous works [56], [60], this study specifically focuses on the experimental validation of SoleSound as a tool for gait analysis.

Unlike similar designs, the system proposed here is fully portable and can estimate a richer set of gait parameters, including inter-limb metrics such as the base of walking. Two calibration strategies were proposed, one focusing on optimizing performance (subject-specific calibration) and one targeting ease of use (generic calibration). Experimental results suggest that SoleSound could be efficiently used as a tool for clinical gait analysis outside the laboratory setting.

In the near future, we will explore applications of SoleSound for Parkinson's patients and elderly individuals with balance disorders.

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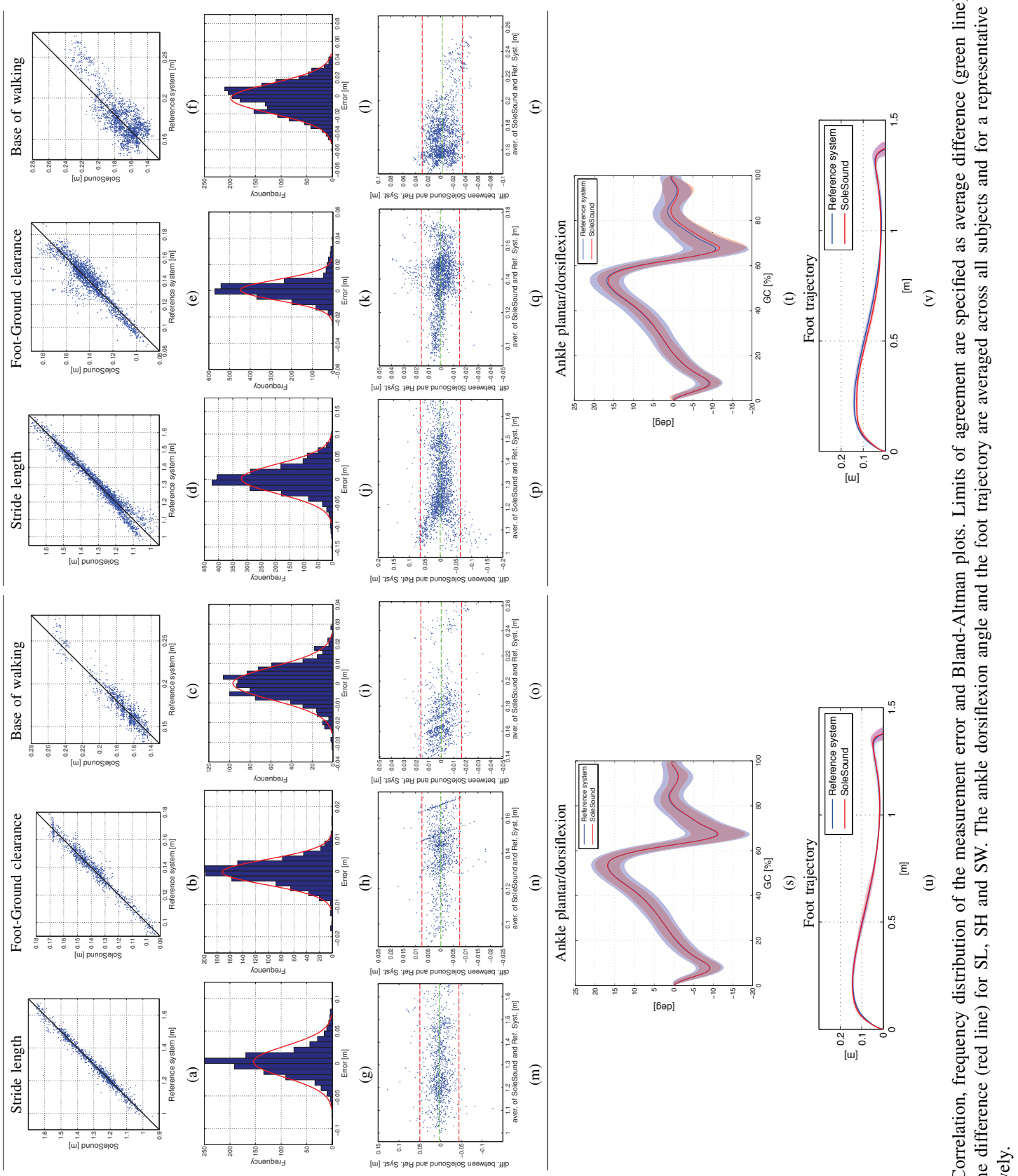


Fig. 5: Correlation, frequency distribution of the measurement error and Bland-Altman plots. Limits of agreement are specified as average difference (green line) ± 1.96 SD of the difference (red line) for SL, SH and SW. The ankle dorsiflexion angle and the foot trajectory are averaged across all subjects and for a representative subject, respectively.

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